

Simple Linear Model

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Model Assumptions and Model Objectives

Consider the general linear model of the form

$$Y = X\beta + \varepsilon \quad (1)$$

where

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \quad X = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix} \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{pmatrix} \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

Here Y is the vector of n observations, β is the vector of parameters, ε is the vector of random errors and X is the design matrix. Both X and β are fixed.

Assumptions

We assume ε to have the following properties:

$$\mathbb{E}(\varepsilon) = 0 \quad (2)$$

$$\text{Var}(\varepsilon) = \sigma^2 I_n \quad (3)$$

The second assumption implies that $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n$ have the same but unknown variances σ^2 and are uncorrelated.

Objectives

- i) To estimate the unknown parameters $\beta_1, \beta_2, \dots, \beta_p$ is possible OR at least estimate those linear combinations of β that can be estimated.
- ii) To test suitable statistical hypotheses about β OR at least functions of β .

Derivation

The rank of X is assumed to r such that $r \leq \min(n, p)$. If $r = p \leq n$, the model (1) is said to be a *full rank model*, otherwise it is called a non full rank model.

Define $\hat{\beta}$ as a function of known quantities like Y, X , etc. such that $\hat{\beta}$ is “close” to β in some sense. Then,

1. $e = Y - X\hat{\beta}$ is the *vector of residuals*.
2. $\varepsilon = Y - X\beta$ is the *vector of errors*.

One way to obtain $\hat{\beta}$ is to use *method of least squares*.

Consider the sum of squares of residuals, i.e.,

$$\begin{aligned} e'e &= (Y - X\hat{\beta})'(Y - X\hat{\beta}) \\ &= Y'Y - Y'X\hat{\beta} - \hat{\beta}'X'Y - \hat{\beta}'X'X\hat{\beta} \\ &= Y'Y - 2\hat{\beta}'X'Y + \hat{\beta}'X'X\hat{\beta} \end{aligned}$$

The last equality follows since $Y'X\hat{\beta} = \hat{\beta}'X'Y$ as both sides are 1×1 matrices.

$$\implies \frac{\partial}{\partial \hat{\beta}}(e'e) = -2X'Y + 2(X'X)\hat{\beta}$$

Equating this to 0 we have,

$$X'Y = (X'X)\hat{\beta} \tag{4}$$

The above collection of equations are called *normal equations*.

When the inverse of $(X'X)$ exists, we simply have

$$\hat{\beta} = (X'X)^{-1}X'Y \tag{5}$$

Some Results

1. The matrix $S = X'X$ is a symmetric matrix having same rank as X .

Proof. If a vector α is orthogonal to the rows of the matrix X , then $X\alpha = 0 \implies X'X\alpha = 0$, i.e., α is orthogonal to the rows of $X'X$ as well.

Conversely, if $X'X\alpha = 0$, then $\alpha'X'X\alpha = 0$ or $Z'Z = 0$ where $Z = X\alpha$. But $Z'Z = z_1^2 + z_2^2 + \dots + z_n^2$ which implies $Z = X\alpha = 0$, i.e., α is orthogonal to the rows of X .

Thus X and $X'X$ has the same rank r . This also shows that the row space of X and row space of $S = X'X$ are the same. $\square$

2. If $\hat{\beta}$ be any particular solution to the (4), then it minimizes $e'e$, the sum of square of residuals.

Proof. Consider any other values of β , say β_0 . Then,

$$\begin{aligned} (Y - X\beta_0)'(Y - X\beta_0) &= (Y - X\hat{\beta} + X\hat{\beta} - X\beta_0)'(Y - X\hat{\beta} + X\hat{\beta} - X\beta_0) \\ &= (Y - X\hat{\beta})'(Y - X\hat{\beta}) + \left\{ X(\hat{\beta} - \beta_0) \right\}'(Y - X\hat{\beta}) \\ &\quad + (Y - X\hat{\beta})' \left\{ X(\hat{\beta} - \beta_0) \right\} + \left\{ X(\hat{\beta} - \beta_0) \right\}' \left\{ X(\hat{\beta} - \beta_0) \right\} \\ &= (Y - X\hat{\beta})'(Y - X\hat{\beta}) + (\hat{\beta} - \beta_0)'X'(Y - X\hat{\beta}) \\ &\quad + (Y - X\hat{\beta})'X(\hat{\beta} - \beta_0) + \left\{ X(\hat{\beta} - \beta_0) \right\}' \left\{ X(\hat{\beta} - \beta_0) \right\} \\ &= (Y - X\hat{\beta})'(Y - X\hat{\beta}) + (\hat{\beta} - \beta_0)'(X'Y - X'X\hat{\beta}) \\ &\quad + (X'Y - X'X\hat{\beta})'(\hat{\beta} - \beta_0) + m'm \quad \text{where } m = X(\hat{\beta} - \beta_0) \\ &= (Y - X\hat{\beta})'(Y - X\hat{\beta}) + m'm \quad \text{(from (4))} \\ &\geq (Y - X\hat{\beta})'(Y - X\hat{\beta}) \quad \square \end{aligned}$$

Thus the minimized value can be unambiguously denoted by $SSE = (Y - X\hat{\beta})'(Y - X\hat{\beta})$ and stands for *sum of squares due to error*.

Generalized Inverse and It's Properties

Definition

1. Any $n \times m$ matrix A^- satisfying the relation $AA^-A = A$ is defined as a *generalized inverse* (*g-inverse*) of the $m \times n$ matrix A .
2. An $n \times m$ matrix A^- is defined to be a *generalized inverse* of the $m \times n$ matrix A if for every u satisfying $\text{rank}(A) = \text{rank}(A|u)$, A^-u is a solution of the equations $Ax = u$ for unknown u . In other words, A^-u is a solution of the equations $Ax = u$ for unknown u whenever the system is *consistent*.

Properties

Define an $n \times n$ matrix $H = A^-A$. Then,

1. $AH = A$
2. $H^2 = H$
3. $\text{Tr}(H) = \text{rank}(H) = \text{rank}(A)$

Proof. The first two properties follow directly from definition.

The first equality in property 3 follows as H is an idempotent matrix (property 2) and is similar to a diagonal matrix containing eigenvalues of H (spectral theorem). But the only eigenvalues of an idempotent matrix are 0 and 1 as the polynomial $x^2 - x$ annihilates H .

For the second equality, observe that $A^-A\alpha = 0 \implies AA^-A\alpha = 0 \implies A\alpha = 0$ and $A\alpha = 0 \implies A^-A\alpha = 0$. i.e, if a vector α is orthogonal to the rows of A , then it is orthogonal to the rows of H and vice versa. Hence $\text{rank } A = \text{rank } H$. $\square$

General Solution for a System of Homogeneous Equations

Observe that, $A(I - H) = A - AH = 0$, by the first property of g-inverse. i.e, each of the n columns $h_1, h_2, \dots, h_n$ of $I - H$ are orthogonal to the rows of A .

Also, $(I - H)^2 = I - H - H + H^2 = I - H$ by second property of g-inverse. i.e, $I - H$ is idempotent.¹ This implies $\text{rank}(I - H) = \text{Tr}(I - H) = \text{Tr}(I) - \text{Tr}(H) = n - r$, where $r = \text{rank}(A) = \text{rank}(H)$ from third property of g-inverse.

Thus, only $(n - r)$ of the columns vectors $h_1, h_2, \dots, h_n$ are linearly independent. Without loss of generality we assume that the first $(n - r)$ columns of $I - H$, $h_1, h_2, \dots, h_{n-r}$ are linearly independent.

Since A is an $m \times n$ matrix of rank r , its rows are n -vectors and therefore we can find at most $(n - r)$ linearly independent vectors orthogonal to them. $h_1, h_2, \dots, h_{n-r}$ is one such set. If there exists any other vector orthogonal to the rows of A , it must be a linear combination of $h_1, h_2, \dots, h_{n-r}$.

Let us now consider a system of homogeneous equations

$$Ax = 0 \tag{6}$$

By definition, x is orthogonal to the rows of A and so any vector satisfying (6) must be a linear combination of $h_1, h_2, \dots, h_{n-r}$. This is equivalent to saying that x will be a linear combination of columns of $I - H$, $h_1, h_2, \dots, h_n$, because $h_{n-r+1}, h_{n-r+2}, \dots, h_n$ are linear combinations of $h_1, h_2, \dots, h_{n-r}$ themselves.

¹Let R be a ring with unity 1_R and e be an idempotent element of this ring. Then, $1_R - e$ is also idempotent.

Therefore, x must be of the form

$$x = z_1 h_1 + z_2 h_2 + \cdots + z_n h_n = (h_1 \ h_2 \ \dots \ h_n)z = (I - H)z \quad (7)$$

for some $z = (z_1 \ z_2 \ \dots \ z_n)'$.²

Conversely, if (7) holds, $Ax = A(I - H)z = (A - AH)z = 0$, by the first property of g-inverse. This shows that the general solution of (6) is given by $\tilde{x} = (I - H)z$.

General Solution for a System of Non-Homogeneous Equations

Consider a consistent system of non-homogeneous equations given by

$$Ax = u \quad (8)$$

If A^- is any g-inverse of A , by definition of A^- , A^-u is a particular solution to (8). Thus $A(x - A^-u) = u - u = 0$ is a system of homogeneous equations in $x - A^-u$. The corresponding general solution of this homogeneous system is $\tilde{x} - A^-u = (I - H)z$ which implies, the general solution to (8) is,

$$\tilde{x} = A^-u + (I - H)z \quad (9)$$

Solution of the Normal Equations (General Form)

The normal equations are given by (as seen from (4)),

$$X'Y = (X'X)\hat{\beta}$$

A particular solution of these equations will be

$$\hat{\beta} = S^-X'Y$$

where S^- is any g-inverse of $S = X'X$. Thus, the general solution of normal equations will be,

$$\tilde{\beta} = \hat{\beta} + (I - H)z \quad (10)$$

where $H = S^-S$ is a $p \times p$ matrix.

Some More Results

3. $X = XH$

Proof. From the first property of g-inverse, we have $SH = S$. Therefore,

$$\begin{aligned} 0 &= (I - H)'(S - SH) \\ &= (I - H)'(X'X - X'XH) \\ &= (I - H)'(X'(X - XH)) \\ &= (X(I - H))'(X - XH) \\ &= (X - XH)'(X - XH) \end{aligned}$$

Note that both the sides of the equality are $p \times p$ matrices. Thus equating the i th diagonal elements of both sides, we get that the sum of squares of the entries in i th row of $(X - XH)$ is zero for every $i = 1, 2, \dots, p$. Thus every element of $(X - XH)$ is zero. $\square$

4. A solution of the normal equations (4) is unique if and only if $\text{rank}(X) = \text{rank}(X'X) = p$.

Proof. A solution to the normal equations is unique if and only if the general solution (10) of normal equations doesn't contain the arbitrary vector.

This happens if and only if $I - H = 0$ i.e., $I = H = S^-S$ which happens if and only if S is non-singular and has regular inverse S^{-1} . Thus the result follows. $\square$

²These z_i 's need not be unique even for a fixed x as columns of $I - H$ are not necessarily linearly independent.

Estimability of a Linear Parametric Function

If $\hat{\beta}$ is a solution to the normal equations (4), then two difficulties arise from the use of $\hat{\beta}$ as estimator of β :

- i) There might exist several solutions for the normal equations (4) in general (recall (7)) and hence $\hat{\beta}$ is usually not unique.
- ii) $\mathbb{E}(\hat{\beta}) = \mathbb{E}(S^-X'Y) = S^-X'X\beta = H\beta$, where $H = S^-S$, $S = X'X$. This implies that $\hat{\beta}$ is not unbiased for β in general.

This compels us to estimate some linear functions of the elements of β , rather than estimating β itself.

Definition

A linear parametric function $\lambda'\beta$ where $\lambda = (\lambda_1 \lambda_2 \dots \lambda_k)'$ is said to be estimable if there exists at least one linear function of observations $u'Y$ where $u = (u_1 u_2 \dots u_n)'$ such that $\mathbb{E}(u'Y)$ is identically equal to $\lambda'\beta$.

By “identically equal to $\lambda'\beta$ ”, we mean equal to $\lambda'\beta$ whatever may be the value of β . We denote this by, $\mathbb{E}(u'Y) = \lambda'\beta$ and now substituting for $\mathbb{E}(Y)$ we have

$$\begin{aligned} u'X\beta &= \lambda'\beta \\ \implies u'X &= \lambda' \end{aligned}$$

The implication follows from the identically equal to condition and hence we get that λ' is a linear combination of the rows of X .

Theorem 1. *A necessary and sufficient condition of estimability of a parametric function $\lambda'\beta$ for the model (1) is $\lambda' = \lambda'H$.*

Proof. The necessary part follows from the following sequence of equalities which are direct consequences of Result 3 and the previous implication.

$$\lambda'H = u'XH = u'X = \lambda'$$

For the sufficient part, if $\lambda'H = \lambda'$, then

$$\lambda' = \lambda'S^-S = \lambda'S^-X'X = u'X$$

where we can take $u' = \lambda'S^-X$. Thus λ' is a linear combination of the rows of X . □

Best Linear Unbiased Estimator (BLUE)

A linear function $b'Y$ of the observations Y in the model $Y = X\beta + \varepsilon$ is said to be the *best linear unbiased estimator (BLUE)* of a parametric function $\lambda'\beta$, if it is unbiased for $\lambda'\beta$ and its variance is the smallest among all linear unbiased estimators of $\lambda'\beta$.

The *Gauss-Markov Theorem* provides an easy method of obtaining the BLUE of any estimable parametric function $\lambda'\beta$ of the above model.

Theorem 2 (Gauss-Markov). *For the model $Y = X\beta + \varepsilon$, $\mathbb{E}(\varepsilon) = 0$ and $\text{Var}(\varepsilon) = \sigma^2I$, where Y is observed, X is known and β, σ^2 are unknown, the BLUE of an estimable linear parametric function $\lambda'\beta$ (with λ known) is $\lambda'\hat{\beta}$, $\hat{\beta}$ being any solution of the corresponding normal equations $X'Y = X'X\beta$, which is obtained by minimizing the quantity $(Y - X\beta)'(Y - X\beta)$ with respect to the unknown vector β .*

Proof.

$$\mathbb{E}(\lambda' \hat{\beta}) = \mathbb{E}(\lambda' S^- X' Y) = \lambda' S^- X' X \beta = \lambda' S^- S \beta = \lambda' H \beta = \lambda' \beta$$

The last equality follows from Theorem 1 due to the estimability of $\lambda' \beta$. This implies that $\lambda' \hat{\beta}$ is an unbiased estimator of $\lambda' \beta$ and is thus eligible for being BLUE.

For $\lambda' \hat{\beta}$ to be BLUE, we have to prove that the variance of $\lambda' \hat{\beta}$ is not larger than any other unbiased estimator of $\lambda' \beta$.

Let $u'Y$ be any other unbiased estimator of $\lambda' \beta$. Then $\mathbb{E}(u'Y) = u'X\beta = \lambda' \beta$ implies that $u'X = \lambda'$ as $u'Y$ is taken to be a unbiased estimator of $\lambda' \beta$.

$$\begin{aligned} \text{Cov}(u'Y - \lambda' \hat{\beta}, \lambda' \hat{\beta}) &= \text{Cov}(u'Y - \lambda' S^- X' Y, \lambda' S^- X' Y) \\ &= \text{Cov}((u' - \lambda' S^- X')Y, (\lambda' S^- X')Y) \\ &= (u' - \lambda' S^- X') \text{Var} Y (\lambda' S^- X')' \\ &= (u' - \lambda' S^- X') X (S^-)' \lambda \sigma^2 & (\text{Var}(Y) = \text{Var}(\varepsilon) = \sigma^2 I) \\ &= (u'X - \lambda' S^- X' X) (S^-)' \lambda \sigma^2 \\ &= (\lambda' - \lambda' H) (S^-)' \lambda \sigma^2 = 0 & (\text{since } \lambda' = \lambda' H) \end{aligned}$$

Using the above and non negativity of variance of a variable, we get the required result.

$$\begin{aligned} u'Y &= (u'Y - \lambda' \hat{\beta}) + \lambda' \hat{\beta} \\ \implies \text{Var}(u'Y) &= \text{Var}(u'Y - \lambda' \hat{\beta}) + \text{Var}(\lambda' \hat{\beta}) + 2 \text{Cov}(u'Y - \lambda' \hat{\beta}, \lambda' \hat{\beta}) \end{aligned} \quad (11)$$

$$\begin{aligned} &= \text{Var}(u'Y - \lambda' \hat{\beta}) + \text{Var}(\lambda' \hat{\beta}) \\ \text{Var}(u'Y) &\geq \text{Var}(\lambda' \hat{\beta}) \end{aligned} \quad (12)$$

□

Remark

Observe from (11) that, equality in (12) holds if and only if $\text{Var}(u'Y - \lambda' \hat{\beta}) = 0$. But we have already seen that $\mathbb{E}(u'Y - \lambda' \hat{\beta}) = \lambda' \beta - \lambda' \beta = 0$. Thus the equality in (12) holds if and only if both expectation and variance of the expression $u'Y - \lambda' \hat{\beta}$ are zero. That is $u'Y \equiv \lambda' \hat{\beta}$ with probability 1.

If $\lambda' \hat{\beta}$ is estimable, $\lambda' \hat{\beta}$ is the BLUE and if any other unbiased estimator of $\lambda' \beta$ has the same variance as $\lambda' \hat{\beta}$, it can not be different from $\lambda' \hat{\beta}$. In other words, BLUE of an estimable parametric function is unique.

Estimation Space

The coefficient vectors of all BLUEs are linear combinations of the columns of X and conversely. The vector space spanned by the columns of X is therefore called the *estimation space*.

Since the rank of X is r , there can be at most r linearly independent estimable functions and BLUEs.

Theorem 3. *For the model $Y = X\beta + \varepsilon$, the BLUE of every estimable parametric function is a linear combination of the left hand side of the normal equations $X'Y = X'X\hat{\beta}$ and conversely, any linear combination of the left hand side of the normal equations is the BLUE of its expected value.*

Proof. For the linear model $Y = X\beta + \varepsilon$ if $\lambda' \beta$ is estimable, its BLUE is $\lambda' \hat{\beta}$ (by Gauss-Markov theorem), which can be written as

$$\lambda' \hat{\beta} = \lambda' S^- X' Y = L' q$$

say, where, $L' = \lambda'S^-$ and $q = X'Y = (q_1 \ q_2 \ \dots \ q_p)'$. The BLUE $\lambda'\hat{\beta}$ is thus a linear combination of the left hand side of the normal equations $X'Y = X'X\hat{\beta}$.

Conversely, if we consider a linear combination $L'q = L_1q_1 + L_2q_2 + \dots + L_pq_p$ of the left hand side of the normal equations, it is the BLUE of its expected values because,

$$\mathbb{E}(L'q) = \mathbb{E}(L'X'Y) = L'X'X\beta$$

and by the Gauss-Markov theorem, the BLUE of $L'X'X\beta$ is $L'X'X\hat{\beta} = L'X'Y = L'q$ where the first equality follows from normal equations. $\square$

Note: $L'X'X\beta$ is estimable as the condition of estimability $L'X'XH = LX'X$ is satisfied as $X = XH$.

Error Space

A linear function of the observations is said to belong to the *error space* if and only if its expected value is identically equal to zero, irrespective of the value of β in the model $Y = X\beta + \varepsilon$.

Theorem 4. *A linear function of observations to the error space if and only if its coefficient vector is orthogonal to the columns of X .*

Proof. From the definition, if $b'Y$ belongs to the error space, then $\mathbb{E}(b'Y) = b'X\beta = 0$. This implies that $b'X = 0$ or equivalently $X'b = 0$. i.e, b is orthogonal to the columns of X .

Conversely, if $X'b = 0$ holds, then, $\mathbb{E}(b'Y) = b'X\beta = 0$ and hence $b'Y$ belongs to the error space. $\square$

Theorem 5. *The coefficient vector of any BLUE (when expressed in terms of observations) is orthogonal to the coefficient vector of any linear function of the observations belonging to the error space.*

Proof. We know that, if $b'Y$ belongs to the error space, b is orthogonal to the columns of X . We also know that, the coefficient vector of any BLUE is a linear combination of the columns of X .

Thus, any vector in the estimation space is orthogonal to any vector in the error space and so we say that the error space is orthogonal to estimation space. $\square$

Theorem 6. *If every BLUE is expressed in terms of the observations Y as $a'Y$, the coefficient vector a is a linear combination of the columns of X and conversely, every linear function $a'Y$ of the observations such that the coefficient vector a is a linear combination of the columns of X is a BLUE of its expected value.*

Proof. If $\lambda'\beta$ is estimable, its BLUE is $\lambda'\hat{\beta} = \lambda'S^-X'Y = a'Y$ (by Gauss-Markov theorem) where $a = X(S^-)'\lambda = Xl$, say with $l = (S^-)'\lambda$. This implies a is a linear combination of the columns of X .

Conversely, if $a = Xl$, then $\mathbb{E}(a'Y) = a'X\beta = l'X'X\beta$. Using Gauss-Markov theorem, normal equations and assumption $a' = l'X'$ respectively, we get that the BLUE of $l'X'X\beta$ is $l'X'X\hat{\beta} = l'X'Y = a'Y$, as required. $\square$

Remarks

- i) Ordinarily, we must check the estimability of $l'X'X\beta$ before applying the Gauss-Markov theorem, but as $a'Y$ is a linear function such that its expected value is $l'X'X\beta$, by definition of estimability, it is estimable.
- ii) The above theorem shows that the coefficient vectors of all BLUE's are linear combinations of columns of X and vice-versa.

- iii) The vector space spanned by the columns of X is thus called the *estimation space*.
- iv) Since the rank of X is r , there are at most r linearly independent estimable functions or BLUEs.
- v) Since the estimation space generated by X has rank r and since we can find $n - r$ linearly independent vectors orthogonal to the columns of X (since every column of X is a n -component vector), the rank of the error space is $n - r$.

Theorem 7. *The covariance between any linear function belonging to the error space and any BLUE is zero.*

Proof. If $b'Y$ belongs to the error space and $\lambda'\hat{\beta}$ is the BLUE of an estimable function $\lambda'\beta$, then

$$\begin{aligned}
\text{Cov}(b'Y, \lambda'\beta) &= \text{Cov}(b'Y, \lambda'S^{-}X'Y) \\
&= b'(\lambda S^{-}X')'\sigma^2 && \text{(as } \text{Var}(Y) = \sigma^2 I) \\
&= b'X(S^{-})'\lambda\sigma^2 = 0 && \text{(from Theorem 4.)}
\end{aligned}$$

□

Projection on the Estimation Space

There is another way of looking at the BLUE of an estimable function $\lambda'\beta$. On account of estimability, there is at least one unbiased estimator $a'Y$ of $\lambda'\beta$. The vector a' can then be split as

$$a' = a'P + a'(I - P)$$

where $P = XS^{-}X'$ is $n \times n$, symmetric and idempotent matrix of rank, $\text{rank}(P) = \text{Tr}(P) = \text{Tr}(S^{-}X'X) = \text{Tr}(H) = r$.

The components $a'P$ and $a'(I - P)$ are orthogonal, since $(a'P)(I - P)'a = a'(P - P^2)a = 0$ as $P^2 = P$ and $P' = P$.